SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: CSA1402 COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO2: Construct *top-down and bottom up* parsers osing *CFG for* parsing conf/icfs *and* error recovery *for* efficient synfax free generation. rsrs

As se ss m ent To ol w e ig h ta g e: 7 0+›

QUESTIONS: Total Marks:50

An n exu re A

SCENARIO BASED TASK

Code of Conduct:

I V.Vamsidhar Reddy 192365029 (Name / Reg No) certify tha t th is sub m is sion is my original work and th a t I ha ve a dhere d to *the guidelin es* sp ecifie d *for this* asses sm en t.
I unders tand tha I a ny viola tion a I ac adem ic integrity rules will result in disc ip linary action.

Signature *of* fhe
sfodenf:
V. Vamsidhar Reddy

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SCENARIO-BASED TASK

Q U ES T IO N S:

1. Sc en ari o: Pars in g T ab ie C o n fl ict in LL(1) W hile de s ignin g an LL(1) p ars e r, a stu de nt co n stru cts a p ars in g ta b le and find s m u ltip le e ntri es in a s in gle cell for a no n-te rm inal and inp ut s ymb o l c o mb in ati o n. T his ie ad s to c o n fu s io n durin g p ars i ng. An alyz e ho w the s ynta x an alyze r us es FI RST and FO LL 0 W s ets in thi s p ro ces s, identify th e rea s o n for the c o n flict, and exp la in ho w it a ffe cts determ inis tic p arsi ng . C o n clu de b y s ugge st in g w h at de cis io n s ho uld b e ta ken to m o dify th e gra m m ar s o that it b ec o m es s uita b ie for LL(1) p ars in g.

2. Sc en ario: Inc orrect Parse Tree G en eratio n A s ynta x an alyzer generates a p arse tre e for th e e xpres s io n id + id id, b ut th e ev alu atio n gives in corre ct res ults du e to im prop er g ro up in g o f op era t o rs. A n al yze ho w the s ynta x an alyz er co ns truc t s th e p ars e tre e, identify th e is s ue rela te d to a s s o ciativ ity, and ex p la in ho w gra m m ar rules in flue nc e th e structure o f the tre e. F in ally, dis c us s w h at dec is io n s ho uld b e ta ken to e n force corre ct a s s o ciativ ity and en s ure a cc urate p arsing .

3. Sc en ario: In v alid Exp re s s io n H and lin g A s ynta x analyz er in a co mp iler rec eiv es th e toke n s eq ue nc e corres p o nding to the e xp res s io n a + b fro m th e l exi cal an al yzer. Alth o ugh all to ke ns are v al id indiv idu ally, th e p ars e r fa ils to g en erate a p ars e tree and rep orts an e rror. In thi s s itu ati o n, an al yze ho w the s ynta x an alyze r identifies th e e rror b a s ed o n gra m m ar rules, d eterm in e th e key is s ue in th e to ke n arran ge me nt, and ex p la in ho w p ars in g tec hniq ue s s uc h a s p red i ctiv e or s hilt-red uc e p ars in g h andle s u ch ca s es. F urth er, dis cu s s w h at de cis io n s ho ul d b e ta ken to imp ro ve e rro r dete ctio n and rec o ve ry w hile ma inta inin g c larity in p ars in g.

4. Sc en ario: Am b ig uity in C o nditio nal S ta te me nts A s ynta x an alyz er is de s ign ed us ing th e gram m ar S . if E th en S I if E th en S else S I a. W hile co nstru ct in g th e p ars in g tab ie, co nflict s arise, and the p arse r cann ot de cide w hi ch p ro du ctio n to ap p l y for cert a in inp uts. In this s ce nario , e xp lain ho w the s yntax an alyze r en co unters a m b ig uity, identify the key is s ue kn o w n a s th e d an glin g els e p ro blem , and dis c us s ho w it imp acts p ars ing. S ug g est an ap p rop riate m o dif ica tio n or p ars in g strateg y to re s o lve th e a mb iguity and en s ure correct a s so ciatio n o f el se cl au s es.

5. Scenario: Operator Precedence ConSict A parser is designed for arithnneGc expressions using th e g ra m m ar E . E + E I E E I id. W hil e p ars in g th e inp ut id + id i d, th e s ynta x an al yzer p ro du ce s an in correct p arse tree d ue to am b ig uity in o p erator prec eden ce. Ex a mi ne ho w the s ynta x an al yzer

RUBRICS FOR CALCULATION:

Scenario based assessment

Criteria	Weightage (&)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20’’’	Demonstrates clear and complete urdcrstardiriqufttic scenario and context	Shows good uridcrstaridinq with rninurqaps	Partial uridcrstaridiriq, misses key aspects	Misiritcrprctsurfailstu understand the scenario
Idcritifieati uriufKcy Issucs	20’’’ 25%	Accurately identifies all rclevaritissucsarid factursinvulvcd concepts effectively to address the scenario	Identifies rnust kty issucs witti minor omissions Applies concepts with minor errors	Idcritificslirnitcd ‹ssuts. somt irrclevaritpuints partially correct application	Urahletuidcritifykcy ‹ssuts
Application of Concepts		Applies appropriate		Limited or	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with SUFWt justifieatiuri	Makcswcak dccisiurswith lirnitcd justifieatiuri	Nuclcardccisiuris rnadcurratiuriale provided
Clarityuf Respons	15’’’	Response is clear, structured, arid logically presented	Generally clear wittirninurissucs	Somewhat uricleuror poorly structured	Disurqarizcdarid difficulttuurdcrstard

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Course:- CSA1402 - Compiler Design

→ CO2 - AT3 - SCENARIO BASED TASK

① Parsing Table conflict in LL(1):

In an LL(1) parser, the parsing table is constructed using the FIRST and FOLLOW set of the grammer.

For a production such as:

$A \rightarrow a$

The parser uses the FIRST(a) set to determine which input symbol can begin the production suppose two production.

$A \rightarrow a$
$A \rightarrow \beta$ is $FIRST(a) \cap FIRST(\beta) = \phi$

For example,

A id +

$A \rightarrow a$ -----

↑

Conflict

Another possible conflict occurs when one production can derive ε. In that case, FIRST and FOLLOW is also satisfy the LL(1) conditions.

(2) Incorrect parse tree generation:

* The syntax analyzer construct a parse by applying production according to the input tokens. For the expression id+id-id. The parse tree determinants the order in which operation are the evaluated.

* If the grammer does not specify operator associativity. For example, It means may be interpreted (id+(id-id)) instead of the correct. It may the incorrect result.

Grammer rules directly influence the structure of the parse tree. A grammer that doesnot enforce associativity allows multiple valid parse tree for the same expression.

Decision; The grammer should enforce left associativity allows multiple for addition and subtraction.

for example,

This ensures expression are evaluated left to right

$E \rightarrow E+T / E-T / T$

$T \rightarrow id.$

(3) Invalid Expression Handling:

* the lexical analyzer correctly identifies the tokens in the expression of a* b as identifier and operators. However, the syntax analyzer checks whether the token follows the grammer rules.

* the error occurs because the grammer expects an operand after the '+' operator but another '*' appears instead. therefore, the tokens sequence violates the grammer.

In predictive parsing (LL), the parser detects the error when no valid production exists in the parsing table for current input symbol. In shift level reduce parsing (LR) the parser detect the error when no valid shift or reduce action.

Decision:-

The parser should implement error detection and recovery techniques such as panic. mode recovery. phase. level recovery or synchronization using Follow sets.

④ Ambiguity in Conditional Statements:

The grammer

* S → if E then S
* S → if E then S else S
* S → a

Is ambiguous because the parser cannot determine which if statement an else belongs to this is known as the dangling else problem.

When constructing the parsing table, conflicts aries because both production begins with same if E then S.

The parser cannot decide wheather to reduce using the first producting or continues parsing the second producing continues.

Decision:-

The grammer should be rewritten to distringush matched and unmatched.

S → M/U

M → if E then M else M/a

U → if E then S / if E then M else U.

⑤ Operator Precedence Conflict.

The grammer;

- $E \rightarrow E+E$
- $E \rightarrow E * E$
- $E \rightarrow id$

It doesnot define operator precedence or associativity.

For an input id + id * id, two different parse trees.

* (id+id) * id
* id + (id * id)

The correct evaluation requires multiplication to have higher precedence the addition. since the grammer,

Decision:- the grammer should be rewritten to enforce precedence

for example, $E \rightarrow E+T/T$

$T \rightarrow T * F/F$

$F \rightarrow id$

Here,

* multiplication has higher precedence than Addition.
* Addition and multiplication are the left associative.